

1) a) Watch the video related to flow visualization ✓

b) Different ways to visualize

- water channel w/ dye injection, surface powder or hydrogen bubbles

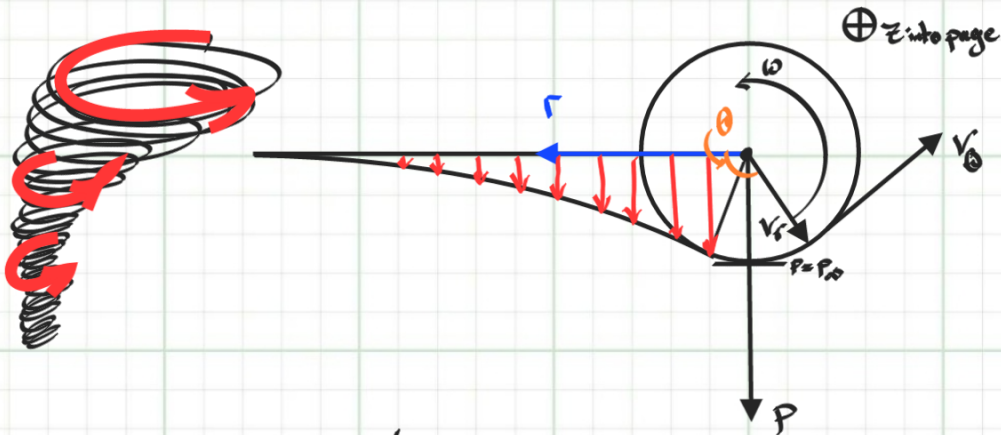
c) Streamlines - as any line at a given instance is everywhere tangent to the velocity vectors of the flow field

Pathline - The trajectory or path of a given small fluid element. a time exposure of fluid element as it travels will give a visualization of the path along that the fluid element travels along.

Streakline - the instantaneous locus or trace of all fluid elements that pass through a given fixed point in space.

d) In steady flow the streakline is the same as the pathline and streamline. Can use either the pathline or streakline to show the streamline pattern.

2)



Approximations

$$V_r = 0$$

$$V_\theta = \begin{cases} \omega r & 0 \leq r \leq R \\ \frac{\omega R^2}{r} & r \geq R \end{cases}$$

Assumptions

2-D motion in $r-\theta$ plane

a) Given Curl (vorticity) of the fluid

$$\nabla \times \vec{V} = \begin{vmatrix} \frac{1}{r} \hat{e}_r & \hat{e}_\theta & \frac{1}{r} \hat{e}_z \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial z} \\ V_r & r V_\theta & V_z \end{vmatrix}$$

$$\Rightarrow \frac{1}{r} \hat{e}_r \left(\frac{\partial V_z}{\partial \theta} - \frac{\partial (r V_\theta)}{\partial z} \right) - \hat{e}_\theta \left(\frac{\partial V_z}{\partial r} - \frac{\partial V_r}{\partial z} \right) + \frac{1}{r} \hat{e}_z \left(\frac{\partial (r V_\theta)}{\partial r} - \frac{\partial V_r}{\partial \theta} \right)$$

$V_z = 0$ "2-D motion" $V_z = 0$ "2-D motion"
 $V_r = 0$ "2-D motion" $V_r = 0$ "2-D motion"
 $V_z = 0$ "2-D motion" $V_z = 0$ "2-D motion"

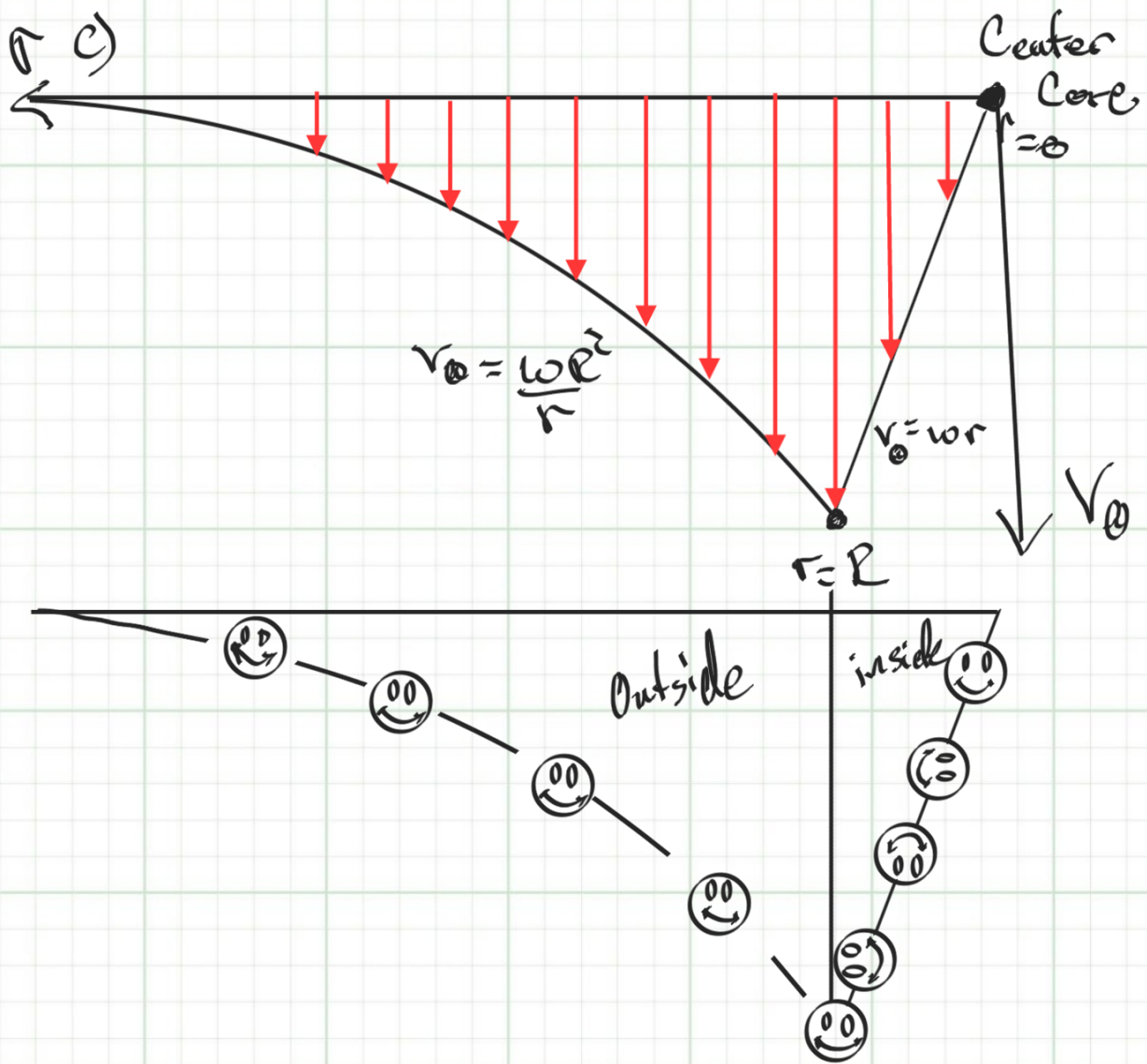
$$\Rightarrow \zeta_z = \frac{1}{r} \left(\frac{\partial (r V_\theta)}{\partial r} - \frac{\partial V_r}{\partial \theta} \right) \Rightarrow \zeta_z = \frac{1}{r} \left(\frac{\partial (r V_\theta)}{\partial r} \right) = \frac{1}{r} \left(\frac{\partial (r \omega R^2 / r)}{\partial r} \right) = \frac{1}{r} \left(\omega R^2 \frac{d}{dr} \right)$$

$\therefore$ Vorticity along the $r-\theta$ plane such that $V_\theta = \frac{\omega R^2}{r}$ for $r \geq R$ equals zero
 $\Rightarrow \zeta_z = 0$ the flow outside the core is irrotational.

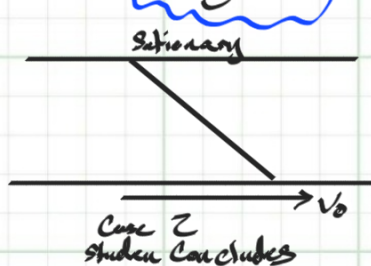
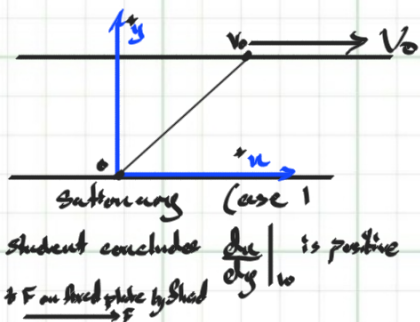
b) given same set up as a) except $V_\theta = \omega r$ for $0 \leq r \leq R$

all conditions still apply thus $\zeta_z = \frac{1}{r} \left(\frac{\partial (r V_\theta)}{\partial r} \right) = \frac{1}{r} \left(\omega r^2 \frac{d}{dr} \right) = \frac{1}{r} (2\omega r)$

$\Rightarrow \zeta_z = 2\omega$ therefore flow inside the core is rotational

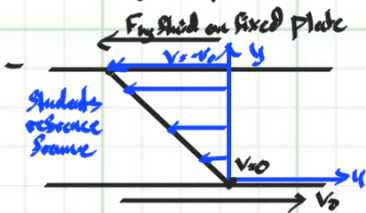


3) Given: $F = (\text{area of plate}) \times \left. \mu \frac{du}{dy} \right|_w$ — Shear Stress in w
on fixed plate



— given no slip condition on top plate will result in fluid motion to the right thus resulting in a force by the fluid which acts on the bottom fixed plate to the right thus the student's analysis for Case 1 is correct

— In Case 1) However is wrong this is because of a mis step in setting of the reference frame the student in this case more than likely choose the bottom plate to be stationary when it is not



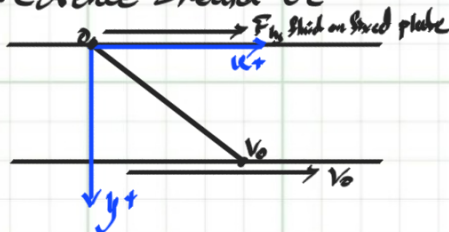
$$V_0 = \frac{du}{dy} \Rightarrow V_0 dy = du$$

$$\text{Since } F = \mu A \left. \frac{du}{dy} \right|_w$$

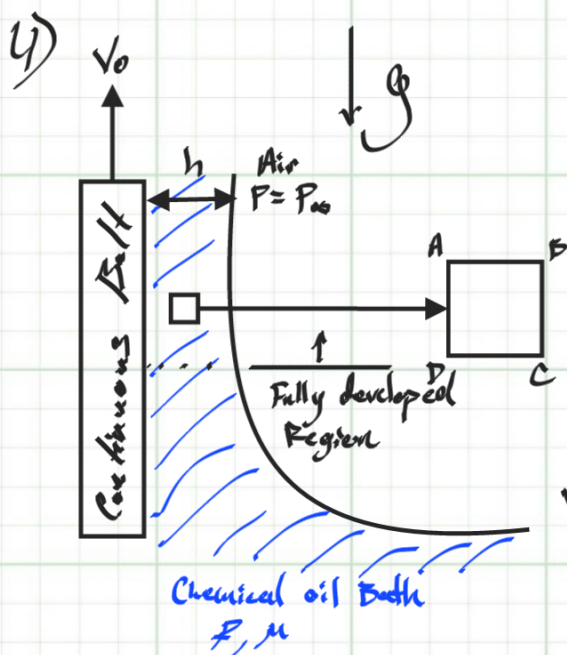
$$\Rightarrow F = \mu A V_0$$

in student's reference frame $\frac{du}{dy}$ is a negative velocity gradient thus

leading them to think the bottom plate to be the fixed point, resulting in a force opposite that of the true velocity gradient of the fluid. Under the same premise as case 1 the frame of reference should be



which is the same as case 1 thus it should result in the same force direction and positive velocity gradient.



using Navier Stokes Eq and Continuity

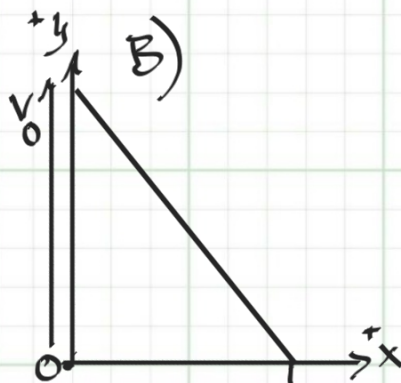
$$\frac{\partial \vec{V}}{\partial t} + \nabla \times (P \vec{V}) = 0 \quad \text{Fully developed region irrotational flow.}$$

Assumptions

$P \rightarrow 0$ negligible $\therefore$ Scale in all directions n

Steady $\rightarrow \frac{d}{dt} = 0$

$u = 0 \therefore$ Continuity $\Rightarrow du = 0$



$y=0$ @ Fully developed region.

A)

x -direction

$$\rho \left(\frac{\partial u}{\partial t} + u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} + w \frac{\partial u}{\partial z} \right) = -\frac{\partial P}{\partial x} + \mu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} \right) + \rho g_x$$

y -direction

$$\rho \left(\frac{\partial v}{\partial t} + u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} + w \frac{\partial v}{\partial z} \right) = -\frac{\partial P}{\partial y} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial z^2} \right) + \rho g_y$$

z -direction

$$\rho \left(\frac{\partial w}{\partial t} + u \frac{\partial w}{\partial x} + v \frac{\partial w}{\partial y} + w \frac{\partial w}{\partial z} \right) = -\frac{\partial P}{\partial z} + \mu \left(\frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} + \frac{\partial^2 w}{\partial z^2} \right) + \rho g_z$$

given $\nabla \cdot \vec{V} = 0$ in x and y all z components of velocities and accelerations $\rightarrow 0$

therefore governing equation to use is

$$\rho v \frac{dv}{dy} = -\frac{dP}{dy} + \mu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) + \rho g_y$$

Since $\frac{\partial v}{\partial y} = 0$ i.e. No change in Velocity components w/ respect to the change in y

$$\Rightarrow \frac{\partial v^2}{\partial y^2} = 0$$

$$\therefore \cancel{\frac{\rho v}{\mu} \frac{dv}{dy}} = \mu \left(\cancel{\frac{\partial^2 v}{\partial x^2}} + \cancel{\frac{\partial^2 v}{\partial y^2}} \right) + \rho g y \quad \text{simplifies further}$$

$$\Rightarrow 0 = \mu \frac{\partial^2 v}{\partial x^2} + \rho g y \Rightarrow g \text{ is } (-)$$

$$\therefore \frac{\rho g}{\mu} = \frac{\partial^2 v}{\partial x^2}$$

$$\frac{\rho g}{\mu} = \frac{d}{dx} \left(\frac{dv}{dx} \right) \quad v \text{ is function of } x$$

by integration

$$\boxed{\frac{\rho g}{\mu} x + C_1 = \frac{dv}{dx}}$$

by subing C_1 back in

$$\frac{\rho g}{\mu} x - \frac{\rho g h}{\mu} = \frac{dv}{dx} \Rightarrow \frac{dv}{dx} = \frac{\rho g}{\mu} (x - h)$$

by integration again

$$\boxed{\frac{\rho g}{\mu} \left(\frac{x^2}{2} - hx \right) + C_2 = v}$$

by Boundary condition

$$v @ x = h \quad \frac{dv}{dx} = 0$$

$$\Rightarrow C_1 = -\frac{\rho g h}{\mu}$$

$$v @ x = 0 \quad v = v_0$$

$$\Rightarrow C_2 = v = v_0$$

$$\therefore \frac{\rho g}{\mu} \left(\frac{x^2}{2} - hx \right) + V_0 = V \Rightarrow V = V_0 - \frac{\rho g}{\mu} (zh - x)x$$

$$D * V_{max} (@ x=0) \Rightarrow V_0 = V_{max}$$

$V_{max} > V_0$ is oil is to flow w/belt

else is $V_{max} > V_0$ there will be No Net oil flow